

Week 10 — Activity: Boundary Conditions and Geological Meaning

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Purpose

To understand how boundary conditions encode geological assumptions in numerical models.

A numerical model does not “know” geology—it only knows equations and boundary conditions.

Boundary conditions determine what kind of Earth your model represents.

Part A: Grounding in real systems

Consider the following settings:

- Ocean floor beneath deep water
- Continental lithosphere
- A hot magma body cooling in the crust

Task: For each case, describe what controls the temperature at the boundaries.

Part B: Types of boundary conditions

Common thermal boundary conditions:

- Fixed temperature (Dirichlet)
- Fixed heat flux (Neumann)

Task: Match each type of boundary condition to a geological situation.

Part C: Comparing assumptions

Explain how the model changes if you prescribe:

- surface temperature
- surface heat flow

What is fixed in each case? What is allowed to vary?

Part D: Geological consequences

Compare appropriate boundary conditions for:

- oceanic lithosphere
- continental lithosphere

Part E: Critical thinking

Task: Explain why inappropriate boundary conditions can produce physically misleading results—even if the numerical solution is mathematically stable.

Reflection

If two models use the same equation but different boundary conditions, can they produce completely different results? Why?

Key Insight

Boundary conditions are where geological assumptions enter a numerical model.